

命题 $Q(m)$: S_n 中任意一个逆序数为 m 的置换都可以表示成 m 个单对换的乘积.

Proof: 我们先证命题 $Q(0)$.

设 σ 是 S_n 中任意一个逆序数为 0 的置换.

$$\because \sigma \in S_n \text{ 且 } l(\sigma) = 0 \quad \therefore \sigma = id$$

$\therefore \sigma$ 可以表示成 0 个单对换的乘积. $\therefore$ 命题 $Q(0)$ 得证.

我们再证命题 $Q(1)$.

设 σ 是 S_n 中任意一个逆序数为 1 的置换.

$$\because \sigma \in S_n \text{ 且 } l(\sigma) = 1 \quad \therefore l(\sigma) \neq 0 \quad \therefore \sigma \neq id$$

$$\because \sigma \in S_n \text{ 且 } \sigma \neq id \quad \therefore \exists k \in \{1, 2, \dots, n-1\}, \text{ s.t. } \sigma(k) > \sigma(k+1)$$

$$\because \sigma \in S_n, s_k = (k \ k+1), k \in \{1, 2, \dots, n-1\}, \sigma(k) > \sigma(k+1)$$

$$\therefore l(\sigma s_k) - l(\sigma) = -1 \quad \therefore l(\sigma s_k) = l(\sigma) - 1 = 1 - 1 = 0$$

$$\therefore \sigma s_k \in S_n \text{ 且 } l(\sigma s_k) = 0 \quad \therefore \sigma s_k = id$$

$$\therefore \sigma = s_k^{-1} = s_k$$

$\therefore \sigma$ 可以表示成 1 个单对换的乘积. 命题 $Q(1)$ 得证.

我们再证命题 $Q(2)$.

设 σ 是 S_n 中任意一个逆序数为 2 的置换

$$\because \sigma \in S_n \text{ 且 } l(\sigma) = 2 \quad \therefore l(\sigma) \neq 0 \quad \therefore \sigma \neq id$$

$$\because \sigma \in S_n \text{ 且 } \sigma \neq id \quad \therefore \exists k \in \{1, 2, \dots, n-1\}, \text{ s.t. } \sigma(k) > \sigma(k+1)$$

$$\because \sigma \in S_n, s_k = (k \ k+1), k \in \{1, 2, \dots, n-1\}, \sigma(k) > \sigma(k+1)$$

$$\therefore l(\sigma s_k) - l(\sigma) = -1 \quad \therefore l(\sigma s_k) = l(\sigma) - 1 = 2 - 1 = 1$$

$$\therefore \sigma s_k \in \mathfrak{S}_n \text{ 且 } l(\sigma s_k) = 1 \quad \therefore \exists \text{ 单对换 } T_1, \text{ s.t. } \sigma s_k = T_1$$

$$\therefore \sigma = T_1 s_k^{-1} = T_1 s_k$$

$\therefore \sigma$ 可以表示成 2 个单对换的乘积. $\therefore$ 命题 Q(2) 得证.

我们再证命题 Q(3)

设 σ 是 $\mathfrak{S}_n$ 中任意一个逆序数为 3 的置换

$$\therefore \sigma \in \mathfrak{S}_n \text{ 且 } l(\sigma) = 3 \quad \therefore l(\sigma) \neq 0 \quad \therefore \sigma \neq \text{id}$$

$$\therefore \sigma \in \mathfrak{S}_n \text{ 且 } \sigma \neq \text{id} \quad \therefore \exists k \in \{1, 2, \dots, n-1\}, \text{ s.t. } \sigma(k) > \sigma(k+1)$$

$$\therefore \sigma \in \mathfrak{S}_n, s_k = (k \ k+1), k \in \{1, 2, \dots, n-1\}, \sigma(k) > \sigma(k+1)$$

$$\therefore l(\sigma s_k) - l(\sigma) = -1, \quad l(\sigma s_k) = l(\sigma) - 1 = 3 - 1 = 2$$

$$\therefore \sigma s_k \in \mathfrak{S}_n \text{ 且 } l(\sigma s_k) = 2 \quad \therefore \exists \text{ 单对换 } T_1, T_2, \text{ s.t. } \sigma s_k = T_1 T_2$$

$$\therefore \sigma = T_1 T_2 s_k^{-1} = T_1 T_2 s_k$$

$\therefore \sigma$ 可以表示成 3 个单对换的乘积 $\therefore$ 命题 Q(3) 得证.

假设命题 Q(λ) 得证 (其中 $\lambda \in \mathbb{Z}_{\geq 3}$). 即: $\mathfrak{S}_n$ 中任意一个逆序数为 λ 的置换都可以表示成 λ 个单对换的乘积

我们现在来证命题 Q($\lambda+1$).

设 σ 是 $\mathfrak{S}_n$ 中任意一个逆序数为 $\lambda+1$ 的置换

$$\therefore \sigma \in \mathfrak{S}_n \text{ 且 } l(\sigma) = \lambda+1 \in \mathbb{Z}_{\geq 4} \quad \therefore l(\sigma) \neq 0 \quad \therefore \sigma \neq \text{id}$$

$$\therefore \sigma \in \mathfrak{S}_n \text{ 且 } \sigma \neq \text{id} \quad \therefore \exists k \in \{1, 2, \dots, n-1\}, \text{ s.t. } \sigma(k) > \sigma(k+1)$$

$$\therefore \sigma \in \mathfrak{S}_n, s_k = (k \ k+1), k \in \{1, 2, \dots, n-1\}, \sigma(k) > \sigma(k+1)$$

$$\because \ell(\sigma s_k) - \ell(\sigma) = -1 \quad \therefore \ell(\sigma s_k) = \ell(\sigma) - 1 = \lambda + 1 - 1 = \lambda$$

$$\because \sigma s_k \in \mathfrak{S}_n \text{ 且 } \ell(\sigma s_k) = \lambda \quad \therefore \exists \text{ 单对换 } T_1, T_2, \dots, T_\lambda, \text{ s.t.}$$

$$\sigma s_k = T_1 T_2 \cdots T_\lambda \quad \therefore \sigma = T_1 T_2 \cdots T_\lambda s_k$$

$\therefore \sigma$ 可以表示成 $\lambda+1$ 个单对换的乘积. 命题 $Q(\lambda+1)$ 得证.

$\therefore$ 对 $\forall m \in \mathbb{Z}_{\geq 0}$, 命题 $Q(m)$ 得证. $\square$

Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, $\mathfrak{S}_n$ 中逆序数最大的置换是唯一的, 就是:

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ n & n-1 & n-2 & \cdots & 2 & 1 \end{pmatrix} \quad \text{最大的逆序数 } \ell(\alpha) = \frac{n(n-1)}{2}$$

Proof: 当 $n=1$ 时, $\mathfrak{S}_1$ 中仅有一个置换 id , $\ell(\text{id}) = 0 = \frac{1 \times (1-1)}{2}$.

结论成立.

对 $\forall n \in \mathbb{Z}_{\geq 2}$. 对 $\forall \sigma \in \mathfrak{S}_n$, 有:

$$\text{Inv}_\sigma = \{ (i, j) \in \mathbb{Z}^2 : 1 \leq i < j \leq n \text{ 且 } \sigma(i) > \sigma(j) \}$$

$$\ell(\sigma) = |\text{Inv}_\sigma|$$

Inv_σ 中的元素个数最多可能为 $\binom{n}{2} = \frac{n(n-1)}{2}$

$$\therefore \text{对 } \forall \sigma \in \mathfrak{S}_n, \ell(\sigma) = |\text{Inv}_\sigma| \leq \frac{n(n-1)}{2}$$

$$\text{考虑 } \alpha = \begin{pmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ n & n-1 & n-2 & \cdots & 2 & 1 \end{pmatrix}$$

对 $\forall (i, j) \in \mathbb{Z}^2$ 且有 $1 \leq i < j \leq n$, 都有: $\alpha(i) > \alpha(j)$.

$$\therefore (i, j) \in \text{Inv}_\alpha \quad \therefore \ell(\alpha) = |\text{Inv}_\alpha| = \binom{n}{2} = \frac{n(n-1)}{2}$$

$\therefore$ 存在 S_n 中的元素 $\alpha = \begin{pmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ n & n-1 & n-2 & \cdots & 2 & 1 \end{pmatrix} \in S_n$, 使得 $\ell(\alpha) = \frac{n(n-1)}{2}$

对 $\forall \beta \in S_n$, 若 $\ell(\beta) = \frac{n(n-1)}{2}$, 则有:

$$\text{Inv}_\beta = \{(i, j) \in \mathbb{Z}^2: 1 \leq i < j \leq n \text{ 且 } \beta(i) > \beta(j)\}, |\text{Inv}_\beta| = \frac{n(n-1)}{2}$$

$$\therefore \text{Inv}_\beta \subseteq \{(i, j) \in \mathbb{Z}^2: 1 \leq i < j \leq n\}.$$

$$\therefore |\text{Inv}_\beta| \leq \left| \{(i, j) \in \mathbb{Z}^2: 1 \leq i < j \leq n\} \right| = \binom{n}{2} = \frac{n(n-1)}{2}$$

$$\therefore \text{Inv}_\beta = \{(i, j) \in \mathbb{Z}^2: 1 \leq i < j \leq n\},$$

$$\therefore \text{对 } \forall (i, j) \in \mathbb{Z}^2 \text{ 且 } 1 \leq i < j \leq n, \text{ 有: } (i, j) \in \text{Inv}_\beta. \therefore \beta(i) > \beta(j)$$

$$\therefore \beta(1) > \beta(2) > \beta(3) > \beta(4) > \cdots > \beta(n-1) > \beta(n)$$

$$\therefore \beta = \begin{pmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ n & n-1 & n-2 & \cdots & 2 & 1 \end{pmatrix} = \alpha. \quad \square$$

$\equiv$
Lemma: $n \in \mathbb{Z}_{\geq 1}$, 对 $\forall \sigma \in S_n$, 若 $\sigma = T_1 \cdots T_\ell$ (其中 $\ell \in \mathbb{Z}_{\geq 0}$ 且 $T_1, \dots, T_\ell \in S_n$ 是单对换), 则有: $\ell \geq \ell(\sigma)$.

Proof: 令 $P_0 = \text{id}$

$$P_1 = P_0 T_1 = T_1$$

$$P_2 = P_1 T_2 = T_1 T_2$$

$$P_3 = P_2 T_3 = T_1 T_2 T_3$$

$$P_{\ell-1} = T_1 \cdots T_{\ell-1} = P_{\ell-2} T_{\ell-1}$$

$$P_\ell = T_1 \cdots T_\ell = \sigma = P_{\ell-1} T_\ell$$

$$\therefore \ell(p_0) = \ell(\text{id}) = 0.$$

~~$$\ell(p_1) = \ell(p_0 T_1)$$~~

$$\ell(p_1) - \ell(p_0) = \ell(p_0 T_1) - \ell(p_0) \leq 1$$

$$\ell(p_2) - \ell(p_1) = \ell(p_1 T_2) - \ell(p_1) \leq 1$$

$$\ell(p_3) - \ell(p_2) = \ell(p_2 T_3) - \ell(p_2) \leq 1$$

...

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$$\ell(p_e) - \ell(p_{e-1}) = \ell(p_{e-1} T_e) - \ell(p_{e-1}) \leq 1$$

$$\therefore \ell(p_e) \leq \ell \quad \therefore \ell(\sigma) \leq \ell \quad \therefore \ell \geq \ell(\sigma) \quad \square$$

定义 (S_n) $n \in \mathbb{Z}_{\geq 1}$, 则有: $S_n := S_{\{1, \dots, n\}} = \{\text{双射 } \sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\}\}$

定义(单对换) 称 $s_i := (i \ i+1) \in S_n$, $i=1, 2, \dots, n-1$ 为 S_n 中的单对换.

Lemma: 对 $\forall \sigma \in S_n$, 若 $\sigma \neq id$, 则 $\exists k \in \{1, 2, \dots, n-1\}$, s.t. $\sigma(k) > \sigma(k+1)$.

Proof: 假设对 $\forall i=1, 2, \dots, n-1$, 都有 $\sigma(i) \leq \sigma(i+1)$.

则有: 若 $\sigma(i) = \sigma(i+1)$, 则 $i = i+1$. 矛盾.

$$\therefore \sigma(i) < \sigma(i+1)$$

$$\therefore \sigma(1) < \sigma(2) < \sigma(3) < \sigma(4) < \dots < \sigma(n-1) < \sigma(n)$$

$$\therefore \sigma(1), \sigma(2), \dots, \sigma(n) \in \{1, 2, \dots, n\}$$

$$\therefore \sigma(1) = 1, \sigma(2) = 2, \dots, \sigma(n) = n \quad \therefore \sigma = id. \text{ 矛盾.}$$

$$\therefore \exists k \in \{1, 2, \dots, n-1\}, \text{ s.t. } \sigma(k) > \sigma(k+1) \quad \square$$

定理 (n 元置换的对换分解) 对 $\forall \sigma \in S_n$, 存在 $l \in \mathbb{Z}_{\geq 0}$ 和一族单对换 $T_1, \dots, T_l \in S_n$ 使得 $\sigma = T_1 \cdots T_l$

当 $l=0$ 时, $T_1 \cdots T_l = id$. 在 σ 的所有单对换分解中, 最短可能的长度为 $l(\sigma)$.
我们称 l 为上述分解的长度

Proof: 对 $\forall \sigma \in S_n$, 有: $l(\sigma) \in \mathbb{Z}_{\geq 0}$.

$\because \sigma$ 是 S_n 中的一个逆序数为 $l(\sigma) \in \mathbb{Z}_{\geq 0}$ 的置换.

$\therefore \sigma$ 可以表示成 $l(\sigma)$ 个单对换的乘积.

若 $\sigma = T_1 \cdots T_\ell$ ($\ell \in \mathbb{Z}_{\geq 0}$ 且 $T_1, \dots, T_\ell \in S_n$ 是单对换), 则有

$$\ell \geq l(\sigma)$$

$\therefore$ 在 σ 的所有单对换分解中, 最短可能的长度为 $l(\sigma)$ $\square$

定义 (n 元置换的既约表法) $\sigma \in S_n$ (其中 $n \in \mathbb{Z}_{\geq 1}$), σ 的长度为 $l(\sigma)$ 的单对换分解称为 σ 的既约表法.

例: $n \in \mathbb{Z}_{\geq 1}$, $m \in \mathbb{Z}_{\geq 1}$ 且 $2 \leq m \leq n$. 考虑

$$\sigma = \begin{pmatrix} 1 & 2 & \cdots & m & m+1 & \cdots & n \\ m & 1 & \cdots & m-1 & m+1 & \cdots & n \end{pmatrix} \in S_n$$

则有: $\sigma = S_{m-1} \cdots S_1$ 且 $l(\sigma) = m-1$.

Proof: $S_{m-1} \cdots S_1 = (m-1 \ m)(m-2 \ m-1) \cdots (2 \ 3)(1 \ 2)$.

$\sigma(1) = m,$	$S_{m-1} \cdots S_1(1) = m$	$\left \begin{array}{l} \sigma(m-1) = m-2, \\ S_{m-1} \cdots S_1(m-1) = m-2 \\ \sigma(m) = m-1 \\ S_{m-1} \cdots S_1(m) = m-1 \\ \sigma(m+1) = m+1 \\ S_{m-1} \cdots S_1(m+1) = m+1 \\ \vdots \\ \sigma(n) = n, S_{m-1} \cdots S_1(n) = n. \\ \therefore \sigma = S_{m-1} \cdots S_1. \end{array} \right.$
$\sigma(2) = 1,$	$S_{m-1} \cdots S_1(2) = 1$	
$\sigma(3) = 2,$	$S_{m-1} \cdots S_1(3) = 2$	
$\sigma(4) = 3,$	$S_{m-1} \cdots S_1(4) = 3$	
$\vdots$	$\vdots$	
$\sigma(m-2) = m-3$	$S_{m-1} \cdots S_1(m-2) = m-3$	

$$\text{Inv}_\sigma = \{(i, j) \in \mathbb{Z}^2 : 1 \leq i < j \leq n \text{ 且 } \sigma(i) > \sigma(j)\}$$

$$= \{(1, 2), (1, 3), \dots, (1, m)\} \quad \therefore l(\sigma) = m-1$$

$\therefore \sigma = s_{m-1} \dots s_1$ 是 σ 的既约表法. $\square$